

MINYANG LI

Google Scholar ◇ Github ◇ Personal Website

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) *Sep. 2023 - Jun. 2027 (expected)*

B.E. in Artificial Intelligence

Current GPA: 3.422/4.3

Selected courses:

- *DSAA2012 Deep Learning: A+*
- *AIAA2205 Introduction to Artificial Intelligence: A+*
- *UCUG1505 Creative Coding and Interactive Art: A+*
- *DSAA1001 Introduction to Data Science and Analytics: A*
- *UFUG1601 Introduction to Computer Science: A*
- *CMAA5025 Computational Techniques for Sketch-based Creativity: A*

National University of Singapore (NUS)

Jun. 2026 - Jul. 2026

Exchange student nominated by HKUST(GZ)

RESEARCH INTERESTS

I am generally interested in visual generative models and their applications (World models, VLA). It is worth mentioning that I am specifically interested in the theories related to diffusion models, as I find them elegant and beautiful.

RESEARCH EXPERIENCE

Topic: High-resolution image synthesis [1]

Apr. 2025 - Jul. 2025

Supervisor: Prof. Ying-Cong Chen

Coworker: Zhen Yang

ENVISON Lab at HKUST(GZ)

- Explored the use of pre-trained diffusion models for generating images at resolutions higher than their training resolution while maintaining visual quality.
- Integrated the proposed method into multiple downstream tasks, including image editing, customized generation, controllable generation, and video generation.
- Conducted qualitative and quantitative experiments to evaluate the method against prior baselines across multiple generation settings.
- Designed and implemented a user study to evaluate perceptual quality and user preferences of generated results.

Topic: Computational hydrology [2]

Jun. 2024 - Sep. 2024

Supervisor: Dr. Qichun Yang

Earth, Ocean and Atmospheric Sciences Thrust at HKUST(GZ)

- Conducted my first research project in computational hydrology, focusing on deep learning methods for geographical spatiotemporal data modeling.
- Preprocessed large-scale geographical spatiotemporal datasets, including resolution alignment, data cleaning, and preparation for model training.
- Configured, trained, and evaluated deep neural networks, with a primary focus on recurrent neural network (RNN)-based models.

- Gained foundational research experience in applying machine learning techniques to hydrological and geospatial prediction tasks.

INTERNSHIPS

AI Researcher (Algorithm Intern)

Mar. 2026 - Sep. 2025

Mentor: Dr. Tianyi Zhang

Knowin AI at Nanshan, Shenzhen, China

- Developed Vision-Language-Action (VLA) models for predicting pick-and-place degrees of freedom from observation images and task prompts.
- Built upon the π_0 -Fast backbone VLM by training a VQ-VAE to replace the original FastTokenizer and fine-tuning the model on customized robotic manipulation datasets.
- Improved the training pipeline by modifying loss functions and supervision strategies, including soft-label targets and self-forcing training to reduce exposure bias from teacher forcing.
- Supported real-world model deployment and testing for robotic pick-and-place tasks.
- Participated in SOP writing and hands-on robotic data collection, gaining practical experience across the full data-model-deployment workflow.

SELECTED PROJECTS

Multi-object image editing

June. 2026 - July. 2025

Self-conducted

In deep learning course

- Developed VGFE, an inversion-free flow-based image editing framework that uses VQA-guided editing strength search to automatically select optimal edit strength while reducing computational overhead. Extended the method to multi-object editing by designing a reassembly and re-editing pipeline tailored for inversion-free diffusion/flow models. (Report)

Skeleton-adhered Style Transfer for Chinese Characters

Oct. 2025 - Dec. 2025

Collaborator: Jian Yang, a PhD student at HKUST(GZ) In CompTec for Sketch-based Creat course

- Developed CalliGen, a sketch-based Chinese calligraphy generation system that transforms rough hand-written skeletons into stylized calligraphic characters while preserving users' structural intent. Built a large-scale paired dataset of Chinese characters and skeletons. Designed an interactive prototype supporting sketch editing, style selection, and real-time generation. (Report)(Code)(Demo video)

Fact-Aware Consistency Scoring for Model-Generated Answers

Apr. 2026 - May. 2026

Collaborator: Zhiyi Chen, Zhiling Li

In Intro2NLP course

- To address the limitations of global embedding similarity in detecting entity, date, number, and span-level errors, proposed FS-BGE, a BGE-M3 ColBERT-based similarity score with factual token reweighting and answer-span mismatch penalties, and further explored bidirectional NLI-based entailment scoring. (Report)(Code)

PUBLICATIONS

- [1] Z. Yang*, G. Shen*, **Minyang Li***, *et al.*, "Rectifiedhr: Enable efficient high-resolution synthesis via energy rectification," in *CVPR Findings*, 2026.
- [2] H. Yu, Q. Yang, L. Yu, X. Li, **Li, Minyang**, and Y. Yang, "Future climate change increases streamflow and risks of hydrological hazards in the pearl river basin," *Water*, 2026.

SKILLS

Programming Languages

Python, A little bit C++ and Javascript

Model toolkit

diffusers, transformers

Machine Learning Tools

Pytorch, Tensorflow, Sklearn, Pandas, Numpy